Formula 1 Podium Prediction Companion

Milestone 1 – analytical report · Kaggle/Ergast dataset (1950–2024) · prediction: *will this driver finish on the podium (positionOrder ≤ 3), using only information known after qualifying and before the race?*

1. Summary

- **Data:** 14 raw tables, kept unchanged; cleaning is done on copies. The modelling table has exactly one row per (race, driver): 24,952 rows for 1950–2024, of which 8,909 (2003–2024) are used for modelling.
- **Features:** 22 leakage-safe features in five groups (start position, driver form, team form, standings before the race, circuit history). A scrambling test shows that no feature changes when the current and all later results are altered.
- **Split (chronological):** train 2003–2019 · validation 2020–21 · test 2022–24 (used once).
- **Models:** a grid-only baseline, logistic regression, gradient boosting and a shallow PyTorch neural network (one hidden layer). Settings were chosen by validation PR-AUC, thresholds by validation F1.
- **Result:** all three models beat the grid-only baseline by a modest margin on the unseen test years, but the gaps between the three are smaller than the test noise (about ±0.03 PR-AUC), so **no model is declared best**. Where a car starts explains most of what the models know.

2. Data audit and cleaning (EDA before and after)

Issue found	Evidence	Action
Placeholders (sentinels)	30 of 120 columns contain `\N` or an empty string (qualifying Q2/Q3 use empty strings)	Both turned into missing values; checked that none remain
Types	23 columns were text (dates, lap times, durations)	Converted to dates, times and numeric seconds; parsed values re-checked against the `milliseconds` columns (0 ms difference for laps and pit stops)
Keys	All primary keys unique and non-null; 23 of 23 foreign keys reconcile (0 orphans)	Nothing to fix; raw files verified byte-identical (md5) at the end
Duplicate driver-race pairs	85 pairs (91 extra rows), all 1950–78: the same driver appears twice in one race (car swaps / shared drives)	Kept the driver's *best recorded position* (see below)
Non-starters	6.4% of rows have no grid slot or never raced; none of them podiumed	Excluded (reconstructed from post-race columns, never used as features); 72 genuine pit-lane starters kept
Text labels	One trailing space (`'Argentinian '`) and the United States spelled both `USA` and `United States`	Stripped and mapped to one spelling; logged

Duplicate rule. When a car is shared, the result is credited to both drivers, so a driver's label should follow his best recorded result in that race. Two drivers who share one car have different driver ids and each keeps his own row; that is why a few 1950–60 races show four or five podium rows. The alternative rule "first result" would lose official podiums (6 podium rows kept against 22 for the chosen rule). Because all affected rows are from before 1979, none of them reaches the 2003–2024 modelling rows or the form features.

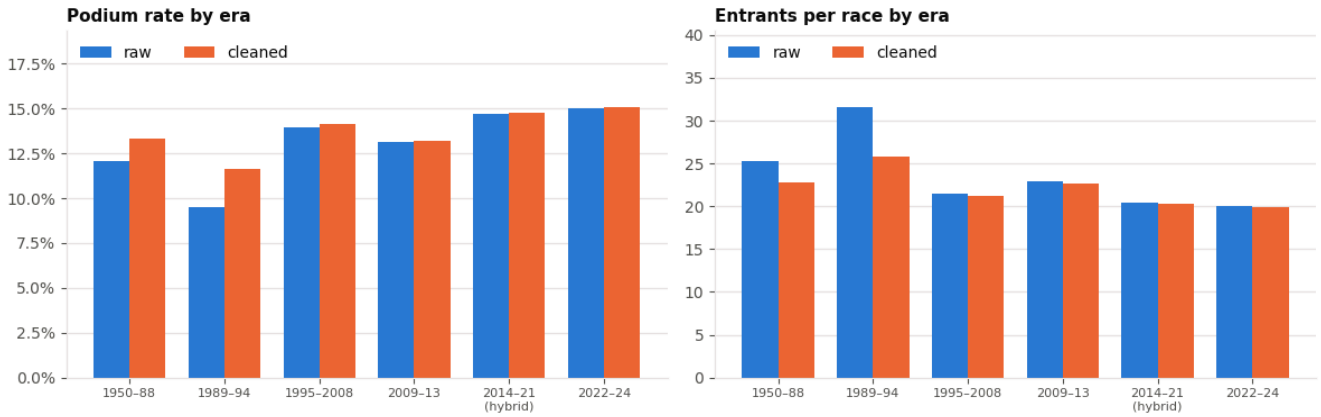


Fig. 1. Podium rate and field size per era, before and after cleaning. Overall podium rate: 12.7% → 13.6%; early eras change most (duplicates and non-starters).

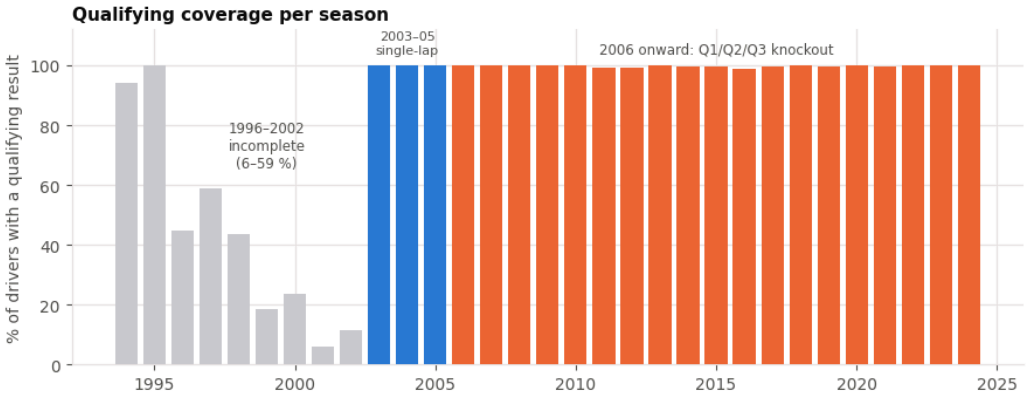


Fig. 2. Share of drivers with a qualifying result per season. No data before 1994, 6–59% in 1996–2002, 99–100% from 2003.

3. Data-engineering questions

Q1. Which circuits convert a front-row start into a podium most often, and did the order change after 2014?

Grid (after penalties) and qualifying position are kept apart: of drivers who qualified on the front row, 815 started there (73% podium) and 36 started further back (31%). Conversion is the share of front-row starts (grid ≤ 2) that end on the podium, for 14 circuits with enough starts in both eras.

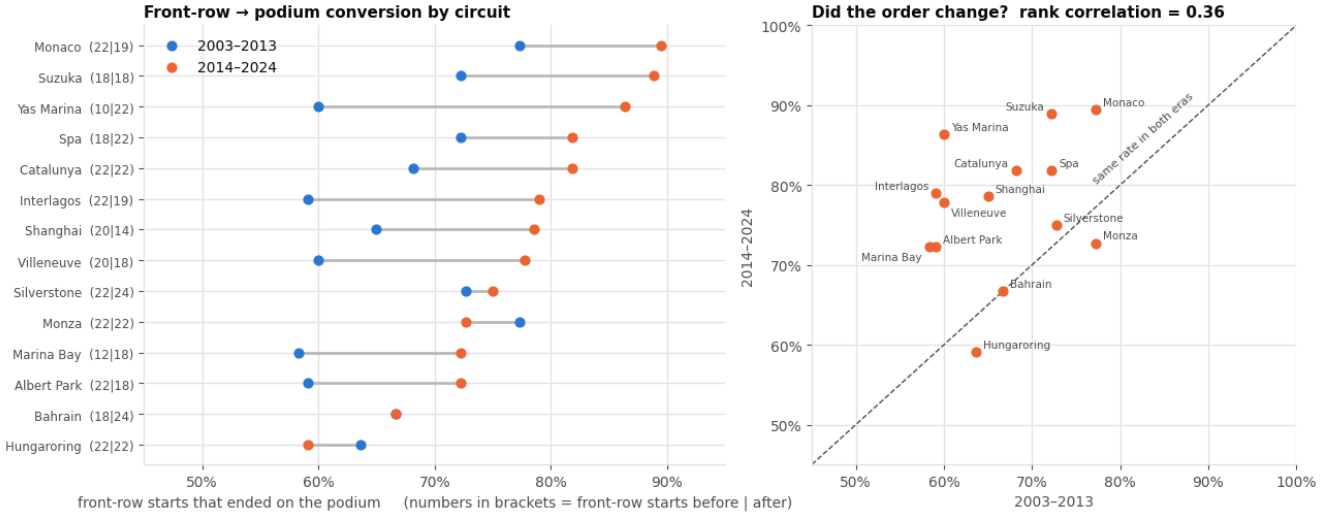


Fig. 3. Front-row conversion per circuit, 2003–13 vs 2014–24, and the rank change between the two eras.

- **Highest (2014–24):** Monaco and Suzuka (89%), Yas Marina (86%). **Lowest:** Hungaroring (59%), Bahrain (67%).

- **After 2014** conversion rose almost everywhere (67% → 77%), but the order changed only weakly (rank correlation 0.36; about −0.1 with older starting years). With about 20 front-row starts per circuit and era, single moves (e.g. Monza −9 places) should not be trusted. What holds in both eras: Monaco and Suzuka high, Hungaroring low.
- **Grain / missingness / cutoff:** one row per (race, driver); qualifying exists for 99.8% of drivers from 2003; 24 circuits with too few starts are not ranked. Grid and qualifying are known before the race; the podium is the outcome, so this analysis is descriptive, and any circuit feature must use earlier races only.

Q2. Do drivers podium more often at home, after controlling for starting position?

Home = driver nationality matches the circuit's country (dual nationalities count in both countries; the Indianapolis 500 and pit-lane starts are removed). Drivers are compared inside the same grid group.

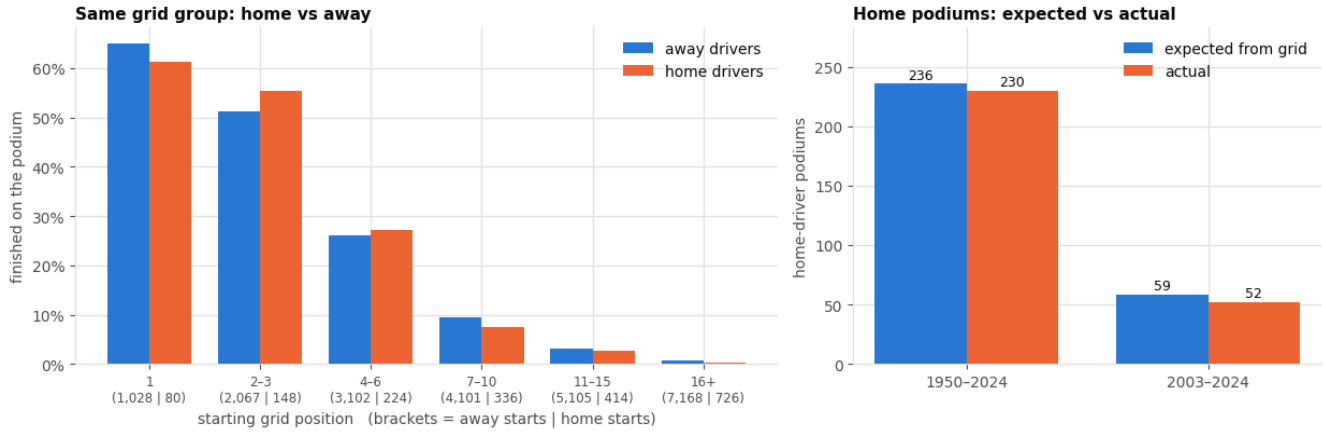


Fig. 4. Podium rate of home and away drivers by grid group (left); home podiums expected from the grid vs actual (right).

- **No.** Raw, home drivers podium slightly less often (about 12% vs 13.9%) because they start further back. Within the same grid group the two are almost identical.
- Home drivers got 230 podiums where their grid slots predict 236 (luck alone moves this by about ±12). In 2003–24: 52 against 59 expected (about ±5). No home advantage is visible.
- **Cutoff / limits:** nationality, circuit country and grid are known before the race. We controlled for grid only, not for car quality; 15 nationalities never have a home circuit in the data.

Q3. Which constructors have the highest mechanical-retirement rate, 2014–21 vs 2022–24?

A mechanical retirement is a race start whose status is a component failure (engine, gearbox, brakes, power unit, …). Crashes, driver errors and regulation decisions are excluded, and three borderline statuses (Retired, Out of fuel, Debris) are not counted but tested. Teams with fewer than 40 starts are not ranked.

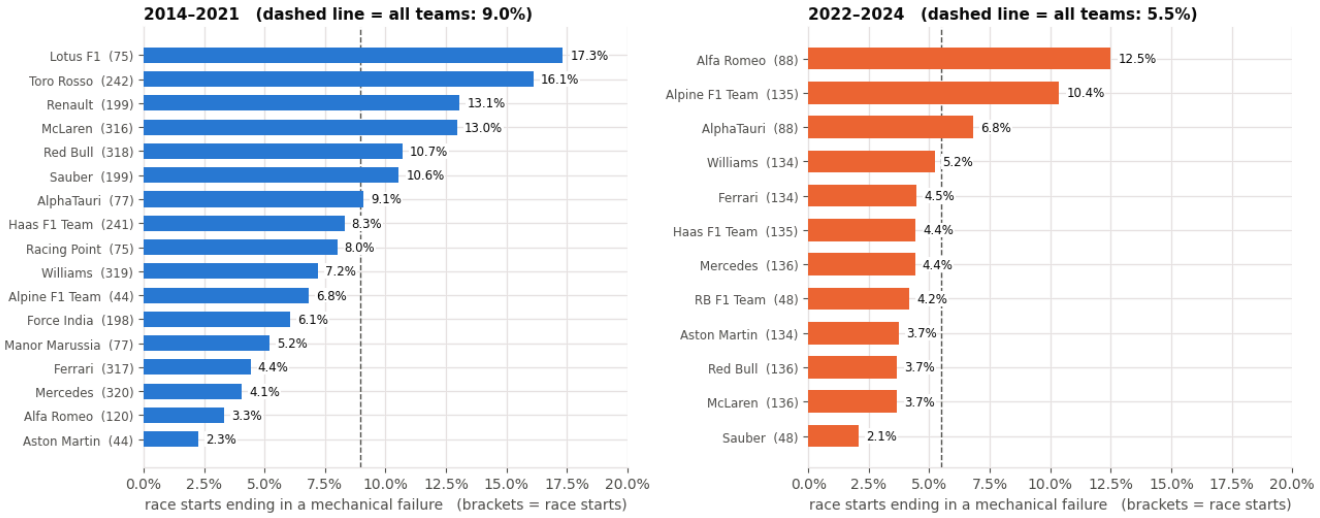


Fig. 5. Mechanical-retirement rate per constructor (dashed line = all teams).

- **2014–21:** highest Lotus F1 (17%), Toro Rosso (16%), Renault and McLaren (13%); lowest Aston Martin (2%), Alfa Romeo (3%), Mercedes and Ferrari (4%).
- **2022–24:** highest Alfa Romeo (12.5%) and Alpine (10%); lowest Sauber (2%), then McLaren, Red Bull and Aston Martin (3.7%).
- Failures became rarer overall (9.0% → 5.5% of starts); McLaren and Red Bull improved most, Alfa Romeo got worse.
- **Caution:** counting the three borderline statuses leaves the 2014–21 top 3 unchanged (rank correlation 0.97) but changes the 2022–24 top 3 (0.63), so the 2022–24 ranking is less stable. Teams have 40–320 starts, so a few failures move a rate a lot; renamed teams are separate rows (each name has its own constructor id).
- **Cutoff:** status is only known after the race, so it can never be a feature.

4. Features and split

4.1 Training years

Qualifying data is the base of the start-position features. It is absent before 1994 and patchy in 1996–2002 (Fig. 2), so training starts in **2003**, the first season with complete coverage; earlier races still feed the form features. The knockout (Q1/Q2/Q3) format only exists from 2006, so for 2003–05 the Q2/Q3 flags are left empty and a `knockout_format` flag tells the model which format applied. We also trained from 1950, 1994, 2003 and 2006: validation PR-AUC differed by less than the validation noise (±0.06) for all models, so the choice rests on data quality.

4.2 Selected features and why each is valid at the pre-race cutoff

Rule: a row may use only what is known after qualifying and before the race, plus results of strictly earlier races. Rolling statistics use `shift(1)`, so the current race is excluded.

Group	Features kept	Why it is available before the race
A. Start position (6)	`grid_pos, quali_pos, gap_pct, reached_q2, reached_q3, knockout_format`	Set by qualifying. `grid_pos` (pit lane = back) and `quali_pos` differ for 27% of drivers (penalties); `gap_pct` = % slower than the fastest time in the same session.
B. Driver form (5)	`drv_podium_5, drv_finish_10, drv_points_5, drv_dnf_5, drv_starts`	Rolling statistics over the driver's previous races only.
C. Team form (4)	`con_podium_3, con_points_5, con_grid_5, con_mech_10`	The team's previous races only (including its mechanical-failure rate).
D. Standings (6)	`drv_st_points/position/wins, con_st_points/position/wins`	Standings after the *previous* round, never after the current one.
E. Circuit (1)	`drv_circ_podiums`	The driver's podiums at this circuit in earlier races.

Dropped: `drv_age` (repeats `drv_starts`, correlation 0.81), `field_size`, `pit_lane_start` and `circ_conv` (no signal, correlation near 0). Window lengths (3, 5, 10 races) give almost the same correlation with the podium (0.50–0.53 for podium rate and team podiums), so one window per feature was fixed.

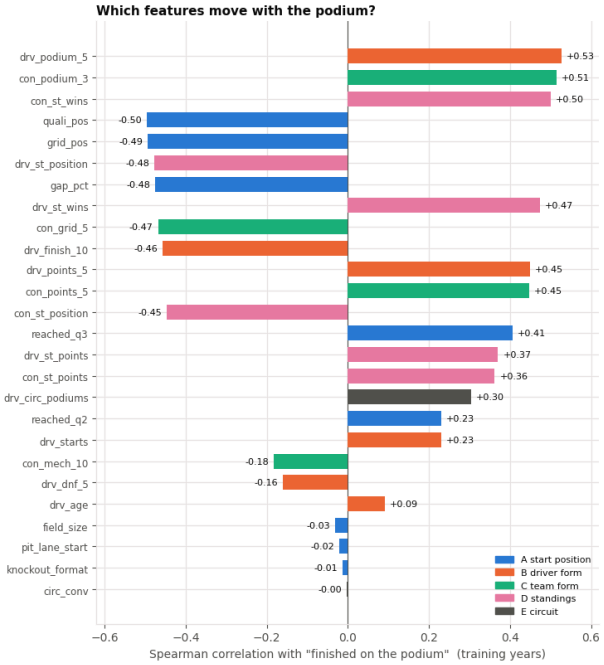


Fig. 6. Correlation of each feature with the podium (training years). Strong ≥ 0.40.

Evidence for the choices

- **Strong (about 0.5):** starting position, recent driver and team podiums, team season wins.
- **Medium:** reached Q2/Q3, season points, driver's podiums at the circuit, previous starts.
- **Weak but kept:** driver DNF rate and team mechanical-failure rate, the only reliability features.
- **Overlap:** many pairs have |correlation| ≥ 0.8 (e.g. grid and qualifying 0.97; driver and team season points 0.95). This is fine for trees and the network; it matters for how credit is split in the explanations (Section 7).
- **`quali_pos` next to `grid_pos`:** logistic regression validation PR-AUC is 0.648 with grid alone, 0.664 with qualifying alone and 0.672 with both. The gain is small, so it is kept as a second description of the start, not because the gain is large.

4.3 Leakage test

For three races (2012 R8, 2019 R12, 2023 R5) the current and all later results were scrambled and all features rebuilt. None of the real features changed, while a deliberately leaky control feature (no `shift`) was detected every time.

4.4 Split

Set	Seasons	Races	Rows	Podium rate	Role
Train	2003–2019	321	6,782	14.2%	fit models, imputers, scalers
Validation	2020–2021	39	775	15.1%	choose settings and thresholds
Test	2022–2024	68	1,352	15.1%	used once, at the end

Why chronological, not random: the features use earlier races, so a random split would put a 2018 race in training and a 2017 race in test (learning from the future) and put drivers of the same race on

both sides, which makes scores look better than they are. A chronological split copies real use. Caveats: 2020 was a shortened season, and 2022 brought new technical rules, so the test is a deliberately harder "new era" check.

5. Models and results

Models. Baseline: logistic regression on `grid_pos` only (the bar to beat). Logistic regression (a weighted sum of the features turned into a probability; easy to read, cannot capture interactions). Gradient boosting (many shallow trees added one after another; captures interactions, can overfit). Shallow neural network (PyTorch, one hidden layer, dropout, early stopping on the newest 15% of training rows, 5 seeds averaged). All use the same 22 features and rows; median filling with a "was missing" flag and scaling are fitted on training rows only.

Metric choice. PR-AUC is primary: podiums are only about 14% of rows, and PR-AUC rewards ranking true podium finishers above the rest without being swamped by the many easy "no podium" rows (ROC-AUC is about 0.91 even for the grid-only baseline, so it separates models poorly). **ROC-AUC is secondary.** F1, precision and recall need a threshold, so they are reported but did not choose settings.

Tuning. A few settings per model were compared on validation PR-AUC; the simplest setting within 0.01 of the best was kept (logistic regression C = 0.1; gradient boosting depth 2, rate 0.03, 300 trees; network 8 hidden units, dropout 0.3). Tuning moved scores by 0.005 at most, so the models are not sensitive to these settings. **Threshold.** Each model's probability becomes "podium" at or above its own threshold, the best F1 on validation (baseline 0.38, logistic regression 0.21, gradient boosting 0.30, network 0.30). The best thresholds are well below 0.5 because podiums are rare.

Model	Set	ROC-AUC	PR-AUC	F1	Precision	Recall
Baseline (grid only)	Train	0.909	0.628	0.648	0.649	0.646
	Validation	0.910	0.648	0.695	0.698	0.692
	Test	**0.913**	**0.647**	**0.649**	**0.650**	**0.647**
Logistic regression	Train	0.933	0.695	0.659	0.535	0.857
	Validation	0.934	0.730	0.717	0.642	0.812
	Test	**0.924**	**0.706**	**0.684**	**0.559**	**0.882**
Gradient boosting	Train	0.945	0.743	0.694	0.610	0.806
	Validation	0.933	0.718	0.733	0.687	0.786
	Test	**0.936**	**0.689**	**0.674**	**0.585**	**0.794**
Neural network	Train	0.935	0.696	0.672	0.567	0.826
	Validation	0.930	0.734	0.712	0.669	0.761
	Test	**0.921**	**0.689**	**0.661**	**0.557**	**0.814**

- **The three models are close to each other**: the test PR-AUC gaps (at most 0.017) are smaller than the test noise (±0.028, bootstrap over races), so the report does not rank them. Logistic regression is used for the worked examples in Section 7 because of a simplicity rule (the simplest model when scores are within noise), decided before looking at test scores.
- **All three beat the grid-only baseline by a modest margin** on PR-AUC (about +0.04 to +0.06). The baseline's own score barely changed between validation and test (0.648 → 0.647), so the gain is not luck of the period. The grid alone already finds most podium finishers.
- **No strong overfitting:** training and validation scores are similar; gradient boosting has the largest train–validation gap. Scores on the test years are usually lower than on validation, as expected for the new rule era.
- **Thresholds are too generous for 2022–24.** On the test rows there are 204 real podium places; the models name 322 (logistic regression), 277 (gradient boosting) and 298 (network) drivers, of whom 180, 162 and 166 are right. Thresholds chosen on 2020–21 do not transfer perfectly to a more competitive era.

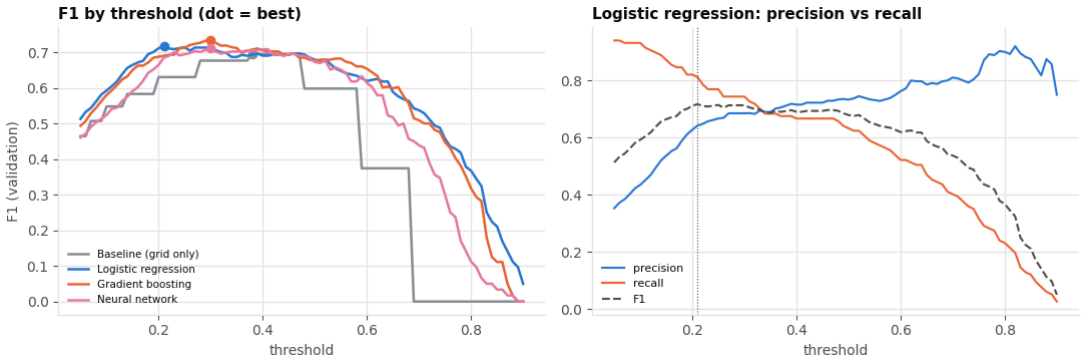


Fig. 7. Validation F1 by threshold for each model (dots = chosen threshold), and precision vs recall for logistic regression.

Example (Abu Dhabi 2024, test). All models named Norris and Sainz (both podium) and wrongly named Verstappen and Piastri (the network also named Russell). Leclerc started 19th and finished 3rd; no model could foresee that from pre-race information.

6. Experiments

6.1 Feature-group ablation (validation PR-AUC lost when a group is removed; tuned models)

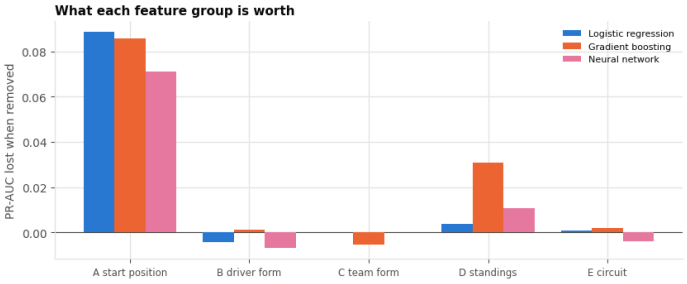


Fig. 8. PR-AUC lost when each group is removed.

Removed	LR	GB	NN
A start position	0.089	0.086	0.071
B driver form	−0.004	0.001	−0.007
C team form	0.000	−0.005	0.000
D standings	0.004	0.031	0.010
E circuit	0.001	0.002	−0.004

All-feature PR-AUC: 0.730 / 0.718 / 0.733. Only A + D together: 0.740 / 0.705 / 0.742.

- **Start position is the only group whose removal clearly hurts** (about 0.07–0.09). Removing driver form, team form or circuit changes little, inside the ±0.06 validation noise, because the groups carry overlapping information. Standings matter a little for gradient boosting.
- **Alone**, each group has signal: start position 0.68–0.69, standings 0.63–0.64, driver form 0.63–0.64, team form 0.53–0.55, circuit 0.24–0.31 (grid-only baseline 0.648).
- "Start position + standings" alone scores about as well as all features, so the history groups add little once the start is known. We keep all groups because they do no harm.

6.2 Preprocessing order (tuned logistic regression and network, validation PR-AUC)

Variant	LR PR-AUC	NN PR-AUC
A split first → fill missing + flag → scale, all fitted on training rows only (ours)	0.731	0.733
B scale first → then fill missing	0.727	0.730
C fill + scale fitted on training + validation rows (information leak)	0.730	0.732
D fill missing, no scaling	0.728	0.724

The order made no practical difference (all within about 0.01, inside the noise). The leaky variant C is not better, so "fit on training rows only" is a matter of principle, not score. Scaling matters more for the neural network (A vs D) than for logistic regression.

7. Explainability

Explanations are not causal. Permutation importance, SHAP and LIME describe what the fitted models rely on, not what makes a driver reach the podium. A feature that is important to a model is not proof that it causes podium finishes. Correlated features (for example team and driver season points) share credit in arbitrary ways, so a push up or down must not be read as "this feature causes a podium".

7.1 Global: permutation importance (validation rows)

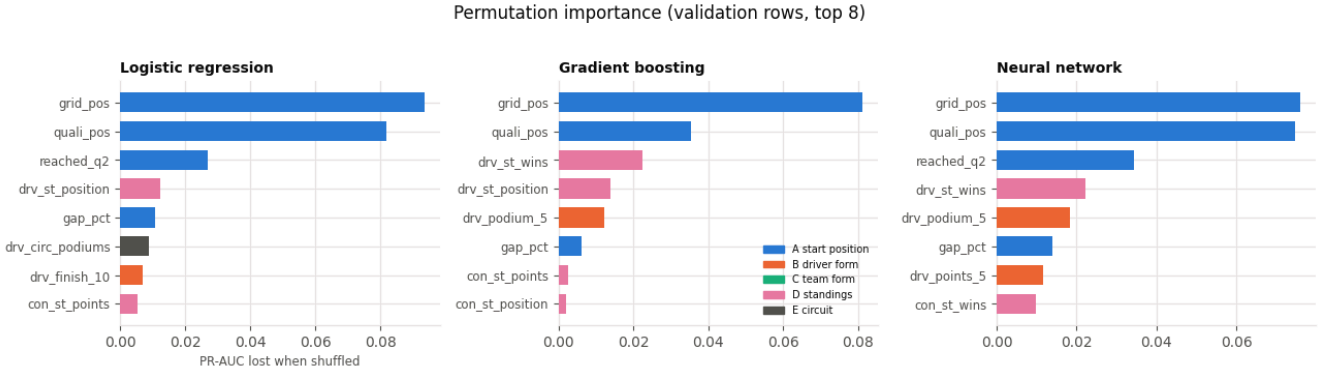


Fig. 9. Permutation importance: PR-AUC lost when a single feature is shuffled (top 8 per model).

- **Start position dominates in all three models:** `grid_pos` loses 0.08–0.09 PR-AUC and `quali_pos` 0.04–0.08 when shuffled alone. Shuffling the whole start-position group costs 0.27–0.33; driver form and standings cost 0.03–0.04; team form and circuit are close to zero, matching the ablation.
- Single-feature numbers understate overlapping features (shuffling `grid_pos` is partly covered by `quali_pos`), so the group view is the cleaner one.

7.2 Global: SHAP (250 validation rows, probability scale)

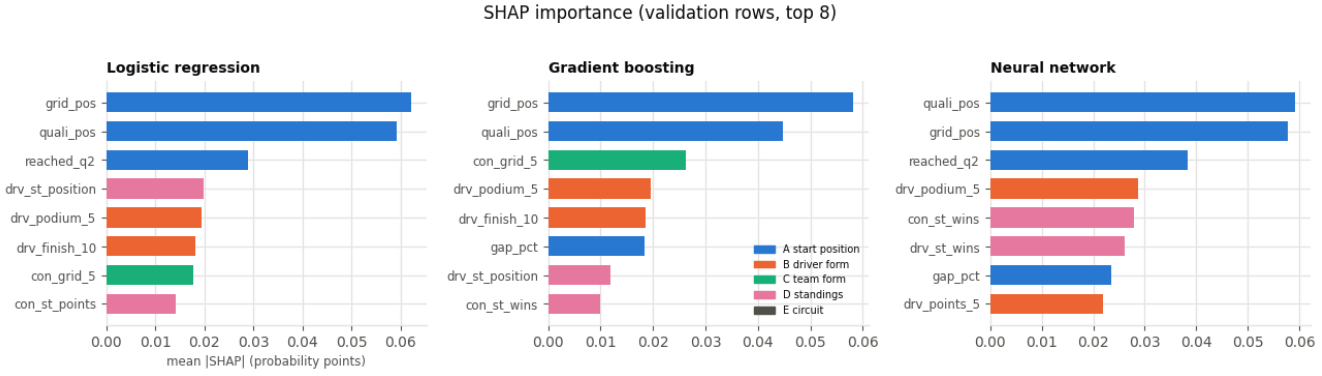


Fig. 10. Mean |SHAP| (average push on the probability) per feature, top 8 per model.

- The same picture: the start-position features lead in every model (about 0.06 probability points each for the top two); the models differ in what comes next (team grid for gradient boosting, season wins for the network).
- In the dot plot (logistic regression) front-of-grid starts push the probability up and back-of-grid starts push it down. `reached_q2` can show a negative push; it overlaps with `quali_pos`, so read it as an overlap effect, not "reaching Q2 hurts".
- Settings: permutation explainer, 100 background rows from the training years, 440 evaluations per row; the neural network is explained with its first seed only.

7.3 Local: two test-race drivers (Abu Dhabi 2024, logistic regression)

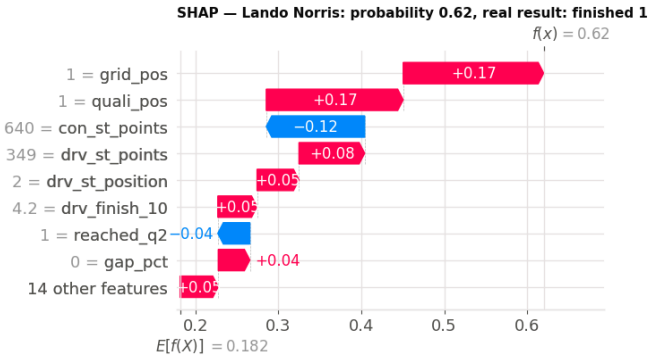


Fig. 11a. SHAP waterfall – Norris (probability 0.62; started 1st, won).

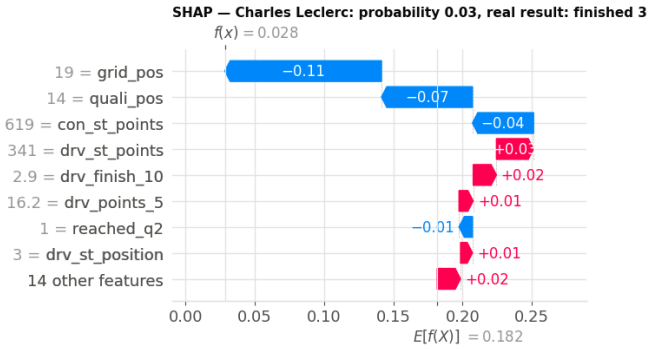


Fig. 11b. SHAP waterfall – Leclerc (probability 0.03; started 19th, finished 3rd).

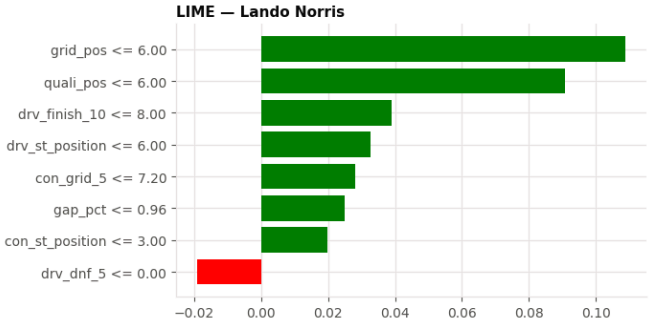


Fig. 12a. LIME – Norris.

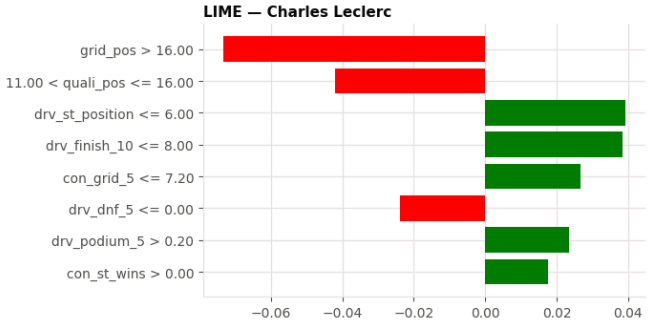


Fig. 12b. LIME – Leclerc.

- **Norris** (0.62 against 0.18 for an average driver): lifted mainly by starting and qualifying 1st (+0.17 each) and by his championship position.
- **Leclerc** (0.03): held down by starting 19th (−0.11) and qualifying 14th (−0.07). His real finish was a recovery from far back, which no pre-race feature can show.
- SHAP and LIME name the same top two features (`grid_pos`, `quali_pos`) with the same direction for both drivers; beyond the top two the lists differ somewhat, as expected for two different methods.
- Oddity: the team's season points push Norris down (−0.12) while his own points push up. The two are 0.95 correlated, so logistic regression splits the credit; read the pair together. This is a property of the model, not a finding about the sport.

8. Inference function

`predict_podium(record)` takes one driver's pre-race record (a dict of the features; any may be left out) and returns the probability and the yes/no answer of each model. Missing values are filled with the training medians (logistic regression, network; a "was missing" flag exists only for features that had gaps in training), and gradient boosting handles gaps natively. Unknown feature names raise an error.

Record (Norris, Abu Dhabi 2024)	Baseline	Logistic reg.	Grad. boosting	Neural net
Complete (all 22 features)	0.68	0.62	0.78	0.53
Partly missing (6 of 22 removed)	0.68	0.93	0.70	0.76
Only the grid slot (21 missing)	0.68	0.18	0.25	0.40

The function answers in all cases, but the probabilities move: **missing information is not neutral** (median filling can push a probability up or down), a limitation to keep in mind.

9. Known limitations

- **Qualifying-coverage drift:** no qualifying data before 1994, 6–59% in 1996–2002, and the Q1/Q2/Q3 format only from 2006 (handled by the 2003 start and the `knockout_format` flag).

- **Era sensitivity:** the training years and the test years are in different technical eras (2022 rule change). Start years made little difference on validation, but the test is a harder check, and the 2022–24 constructor ranking (Q3) is less stable than 2014–21.
- **Thresholds** chosen on 2020–21 are too generous on 2022–24: the models name more drivers than there are podium places.
- **Excluded rows:** non-starters, and pit-lane starters who retired on lap 1 (they look like non-starters in the data); a handful of rows. Duplicate-rule choices affect 1950–78 only.
- **Small samples:** validation and test are small, so differences between models are within noise and no model is declared best.
- **Missing data is not neutral** in the inference function (Section 8).
- **Home-advantage and reliability answers** control for grid only, not for car quality; small team counts make single rates unstable.
- **Explanations are not causal** (Section 7).

10. Reproducibility and references

One Run-All Kaggle notebook (sections: audit, cleaning, Q1–Q3, features and split, training, experiments, test, explainability, inference, limitations). Raw files stay unchanged (md5-verified); the modelling table, predictions and metric tables are saved as CSV. Random seeds are fixed. References: Kaggle "Formula 1 World Championship (1950–2020)" (Rohan Rao; Ergast data) and the course primer *Introduction for Formula 1 Beginners*; scikit-learn (Pedregosa et al., 2011); PyTorch (Paszke et al., 2019); SHAP (Lundberg & Lee, NeurIPS 2017); LIME (Ribeiro, Singh & Guestrin, KDD 2016).